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A palaeomagnetic study of the intrusions and Carboniferous sediments at Dunbar, Scotland

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Synopsis

Gently tilted Lower Carboniferous sediments on the Dunbar foreshore are intruded by a 20 m thick dyke and a subparallel narrow dyke. Heating by the Big Dyke can be traced for 80 m, but the strongly altered Small Dyke only has a 1 m aureole. The Big Dyke remanence differs from that of the unheated sediments, which show similar directions to the Small Dyke. The sequence of superimposed components in the aureole of the Small Dyke suggest that the remanence of the unheated sediments is oldest, followed by the baking effect of the Big Dyke and then by the Small Dyke. This implies a Lower Carboniferous age for the magnetization of the sediments and the Big Dyke, and a Permian age for the Small Dyke. However, the corresponding pole positions suggest Permian ages for the sediments and Small Dyke, and Lower Carboniferous for the Big Dyke. Only the reversed pole (23°S 341°E) for the Big Dyke appears to be substantiated.

Introduction—geology and sampling

Sedimentary rocks of Upper Old Red Sandstone and Lower Carboniferous age have been penetrated by volcanic necks and related intrusions in the Dunbar region. The red sandstones have not yet yielded fossils (Francis 1962), but are overlain by the Cementstone Group of the Calciferous Sandstone Measures of Lower Carboniferous age. In the area of the Belhaven shoreline (56·0°N 2·5°W) the Cementstone Group comprises a series of rhythmic alternations of sandstones and mudstones dipping at shallow angles ($0 - 10°$) to the SE. These have been cut by volcanic tuffs and breccias and then intruded by quartz-dolerite dykes aligned approximately ENE. The widest of these is some 20 m wide (the "Big Dyke"), while a narrower one, *c.* 0·4 m wide (the "Small" Dyke), lies immediately south and at an angle of *c.* 20° to the Big Dyke (Fig. 1). The Big Dyke appears fresh, but the Small Dyke comprises haematite-impregnated, argillised (partially converted to clays), carbonated basaltic rock, with closer affinities to the olivine basalt suite rather than to quartz dolerites (A. Davies, B.G.S., pers. comm.). The age of these intrusions has been suggested as probably late Carboniferous (Francis 1962), but a Permian age cannot be excluded on the geological evidence.

Field drilling yielded at least six separately oriented cores from each site; each core oriented with a sun compass. The initial study was of the Big Dyke on the

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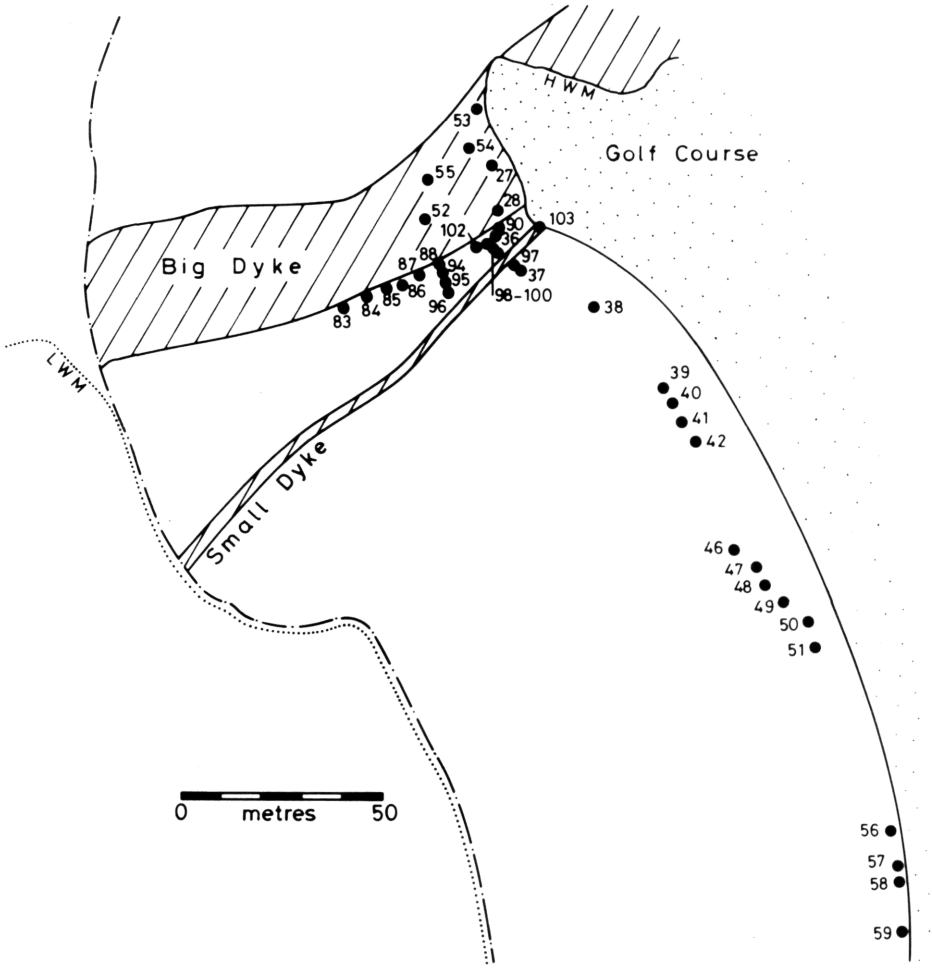


FIG. 1. Sampling sites on the Dunbar Beach.

Belhaven Beach (6 sites) and sediments further south (19 sites). As the extent of baking by the intrusion became evident, further sites were sampled in the zone south of the Big Dyke (Fig. 1). (Fuller details are given by Xu, 1984).

THE PALAEOMAGNETIC PROPERTIES

The Big Dyke

The intensity of natural remanence varied considerably (28 to 2926 mA/m) but the initial directions at five of the six sites were mutually consistent, southerly and

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shallow, while site 52 had directions close to the present geomagnetic field (Table 1). A distinct Curie temperature of 580°C (magnetite) was determined for each site, with an additional, very weak Curie temperature between 300 and 400°C. On cooling in the Curie balance, the saturation intensity decreased slightly, suggesting that a small fraction of magnetite had oxidized to haematite. One pilot specimen from each site was thermally demagnetized in broad increments. Each showed a marked drop of intensity between 460 and 560°C. Another pilot specimen was then chosen from each site in order to study the detailed behaviour of the remanence between 500 and 650°C, close to the Curie temperature (Fig. 2a). In these samples, the intensity drop was sharp between 550 and 580°C, so that their maximum blocking temperatures correspond to their Curie temperatures. Both removed and remaining components, in this temperature range, had identical directions, although the vector removed was larger and hence better defined. The mean of the "most stable" direction (Tarling and Symons 1967) was 196°, 16° ($\alpha_{95} = 10°$) and the mean direction of the linear vectors isolated between 530 and 560°C was 192°, 19° ($\alpha_{95} = 10°$). All remaining specimens were then subjected to thermal demagnetization at the optimum temperature indicated by the pilot specimen from each site. The corresponding mean direction of these sites was 196°, 19° ($\alpha_{95} = 12°$). The Big Dyke is thus characterized by a single component, mostly in the range 250 to 580°C which has a direction, uncorrected for tectonic tilt, of 196°, 19° ($\alpha_{95} = 12°$).

The Unbaked Sediments

The field identification of sediments which had been thermally affected by the intrusions, other than those immediately adjacent to the dykes, was not possible and so the distinction between partially baked and unbaked sediments was determined *a posteriori* i.e. using palaeomagnetism to identify sites with magnetic components attributable to heating by the intrusions. Unaffected sediments were only unambiguously identified on a regional scale in 12 sites (44–51, 56–59) which were more than 80 m south of the Big Dyke (Fig. 1). In these sites, the initial intensity ranged between 1·2 and 30·5 mA/m and the directions were all southerly with medium, upward inclinations. Their Curie points of 670°C and isothermal analyses suggested that haematite was the dominant magnetic mineral, although the saturation intensity only decreased gradually to zero at 670°C.

One or more pilot specimens from each of the 12 sites was thermally demagnetized up to 700°C (Table 1). All showed identical behaviour (e.g. Fig. 2b), with an antiparallel component removed by 55°C, and a single component remanence from at least 200 to 560°C, followed by a decrease in intensity to 550–650°C generally accompanied by an increase (*c.* 20%) in susceptibility. The linear vector between 150 and 560°C had a mean direction 194°, −43° ($\alpha_{95} = 9°$), identical to that defined by the mean of their most stable directions 197°, −43°

TABLE 1

Site mean directions and most stable directions—Big Dyke and unmetamorphosed sediments.

	Demagnetized						Stability				Linearity					
											°C 650–530		°C 530–560		°C 560–580	
Site	Int (mA/m)	Tmt	Int (mA/m)	Dec.	Inc.	α_{95}	Dec.	Inc.	Range	S.I.	Dec.	Inc.	Dec.	Inc.	Dec.	Inc.
Big Dyke																
27	1914	150	1789	190	25	6	186	17	100–460	12						
							196	22	400–430	11	199	27	194	22	229	54
28	20	500	4	197	31	7	178	33	100–260	3						
							185	27	20–300	6	186	30	—	—	—	—
52	26	440	11	188	32	13	53	77	20–100	2						
							185	42	440–530	3	180	24	209	31	203	15
53	2668	340	3804	196	15	8	202	22	340–550	10						
							190	13	340–500	5	188	9	188	3	197	16
54	2736	340	3600	199	11	11	184	14	270–440	11						
							197	13	100–550	7	199	12	186	18	210	12
55	1297	340	1020	203	−2	11	209	2	175–340	9						
							211	2	200–400	3	320	−66	200	12	250	13

Unmetamorphosed sediments											Linear Vectors			
											Dec	Inc	Range	diag. angle
44	2	150	2	202	−35	17	186	−36	55–150	6	150	−11	260–690	7·8
45	17	150	15	193	−51	8	198	−42	55–200	12	204	−49	55–zero	3·3
46	12	150	12	193	−45	4	195	−47	55–160	40	193	−44	200–690	3·5
47	17	150	17	190	−50	4	192	−49	150–460	14	192	−48	200–zero	1·8
48	8	150	8	202	−54	8	203	−27	55–460	10	202	−27	200–zero	2·6
49	8	270	6	197	−44	9	201	−51	200–340	9	201	−50	200–560	2·4
50	3	150	3	199	−48	3	192	−42	100–200	6	195	−46	200–340	5·2
51	21	150	18	196	−50	6	200	−47	100–200	19	197	−51	260–560	3·0
56	10	270	9	200	−33	5	205	−26	175–440	9	204	−29	340–zero	2·4
57	26	350	17	204	−55	3	200	−52	270–460	13	201	−52	340–zero	1·4
							201	−51	475–560	5	201	−48	520–585	3·1
58	18	350	14	189	−47	3	200	−47	50–440	11	197	−47	340–zero	1·2
59	2	350	15	189	−37	6	192	−46	50–440	12	192	−47	175–zero	2·4

The directions of remanence (dec. inc.) in the Big Dyke are uncorrected for geological tilt, but the directions in the sediments are bedding corrected, although this is small, less then 12°. The treatment is always thermal (°C). S.I. is the stability index of Tarling & Symons (1967) in which values greater than 2·5 are considered stable. The range is that in which the least scatter occurs, and is given in °C. Linearity is as defined by Kirschvink (1980) and no lineation is considered to be present if the diagonal angle (diag. angle) is ≥10°.

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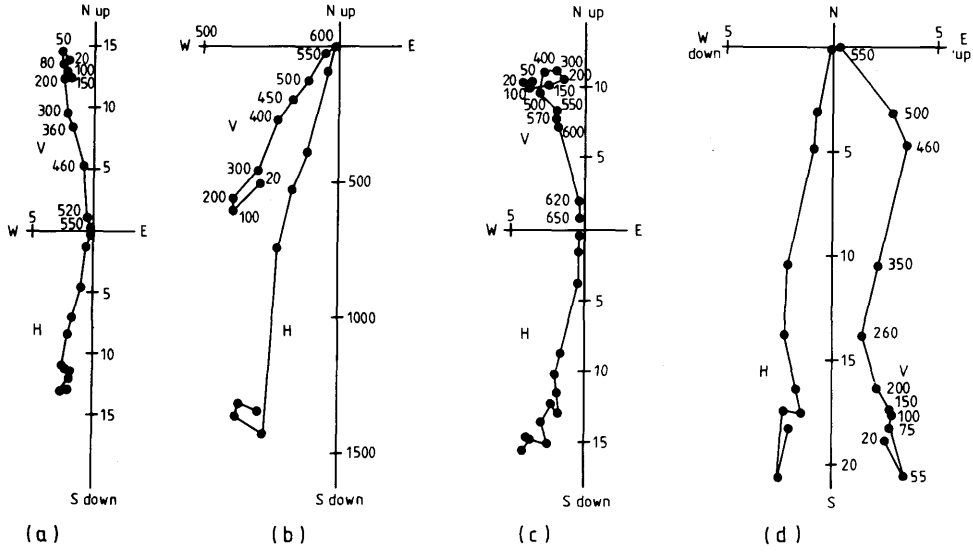


FIG. 2. Cartesian projections of the remanence during thermal demagnetization.
Examples are given for the behaviour of specimens from (a) the unmetamorphosed sediments, (b) the Big Dyke, (c) the baked sediments adjacent to the Small Dyke, and (d) sediments within the Big Dyke baked zone.
Note that the intensity scales change from one plot to another, although they are constant for each plot. H corresponds to the horizontal component, and V to the vertical component. This method of projection was originally used for palaeomagnetic purposes by As (1960). The directions of the axes for plot d are not the same as for a, b and c.

($\alpha_{95} = 5°$). All remaining specimens were then heated to between 150 and 350°C, depending on the behaviour of the appropriate pilot specimen (Table 1). The mean site direction, after this treatment, was 196°, −46° ($\alpha_{95} = 4°$). All three analyses therefore indicate the same characteristic direction for the unmetamorphosed sediments (corrected for shallow bedding) of about 194°, −46°.

The Small Dyke and its Aureole

Pilot specimens from the dyke itself (Site 103) showed a very well defined characteristic single component magnetization (Fig. 2c) between 20 and 650°C, best defined between 450 and 550°C as 189°, −33° ($\alpha_{95} = 1°$). Isothermal remanent studies also indicated haematite, but major changes in the magnetic mineralogy occurred above 550°C when the susceptibility increased threefold. Specimens from within a few centimeters of the dyke showed three components of magnetization (Fig. 2d); (i) an "unmetamorphosed" component of 193°, −40° between 460 and 650°C, (ii) a Big Dyke component of 190°, 13° between 260 and 460°C, and (iii) a

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Small Dyke component of 168°, −19° between 150 and 260°C. A further 6 pilot specimens, one from each site within 1 m of the Small Dyke, also showed an apparent Small Dyke component at temperatures below 200°C, but this became indistinguishable from viscous components below 100°C.

The Metamorphosed Sediments

Within 1 m of the Big Dyke, 7 sites (83–88, 90) showed initial intensities between 12 and 99 mA/m, consistent with having been baked. One or two pilot specimens from each site were thermally demagnetized. Above 55°C, all specimens from near to the dyle showed a clearly linear, stable component, although varying somewhat in direction (Table 2). The mean direction of these linear components was 189°, −7° ($\alpha_{95} = 15°$), i.e. between that of the Big Dyke and the unmetamorphosed sediments. There is no clear difference in the field relationships between these sites, but four site directions can to be readily related to the Big Dyke, having a mean direction of their linear vectors of 185°, 5° ($\alpha_{95} = 18°$), while the other 3 sites had a mean direction of 194°, −23° ($\alpha_{95} = 9°$), nearer to the unmetamorphosed sediment direction than to that of the Big Dyke.

At increasing distances from the Big Dyke, the extent of its heating was variable—some sites exhibiting clear linear, stable components similar to that of the Big Dyke, but most showing unmetamorphosed sediment directions except very near to the Small Dyke (Table 2). Three sites (100–102) between 1 and 3 m of the Big Dyke had linear components (mean direction 197°, −32°, $\alpha_{95} = 24°$) more related to the unmetamorphosed sediments than the Big Dyke, while at greater distances from it, a Big Dyke component was still identifiable within the zone between the Big and Small Dykes (9 sites, 186°, 13°, $\alpha_{95} = 11°$), while only one site (95) had an unmetamorphosed sediment vector (185°, −24°). On the southern side of the Small Dyke and north of the unmetamorphosed zone, a Big Dyke component could be found between 150 and 460°C at only 2 sites (190°, 16°, $\alpha_{95} = 17°$), while unmetamorphosed sediment directions were found at higher temperatures (greater than 350°C) in 4 sites (192°, −33°, $\alpha_{95} = 6°$). Sites 38 to 43 did not show very clear stability and were characterized by components that appeared to be trending towards the present geomagnetic field direction (except site 41).

Interpretation

The extent of the thermal effect of the Big Dyke is somewhat surprising, extending for some 80 m. Although the Big Dyke is large, such heating suggests slow cooling such as would occur if the sediments were then deeply buried. This seems unlikely as other sediments show no sign of a burial magnetization and Francis (1975) considered that the intrusion was into wet sediments. It seems more

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TABLE 2

Linear magnetic components within the baked sediments at temperatures ≥200°C.

Site	Range (°C)	Dec	Inc
Big Dyke contact (≤1 m)			
88	150–zero	189	−18
84	450–560	165	12
85	250–zero	193	−26
86	250–zero	199	−25
87	200–650	193	9
88	250–zero	188	8
90	150–650	193	−8
Metamorphosed zone (>1 m)			
102	300–400	195	−27
101	600–675	205	−49
100	550–650	194	−21
94	350–650	204	16
99	250–650	188	22
36·1	340–560	190	8
36·11	250–550	191	14
95	550–zero	185	−24
96	250–500	204	13
98	200–300	138	14
97	350–zero	199	−41
37·1	460–675	188	−30
37·10	550–675	195	−34
38	340–675	254	83
39	—	—	
40	20–630	330	12
41	55–zero	183	−32
42	20–675	331	53
43	200–zero	311	29
	100–200	18	−47
	300–400	187	17
	250–400	187	7
	200–300	188	4
	250–550	182	11
	100–200	304	−3
	150–250	186	−27
	260–460	191	18
	250–350	182	23
	150–340	195	6

The linearity is as defined by Kirschvink (1980)—see legend to Table 1. The linearity range is in °C, but zero corresponds to the inclusion of the origin of the cartesian coordinates, i.e. beyond the last heating, i.e. NOT 0°C.

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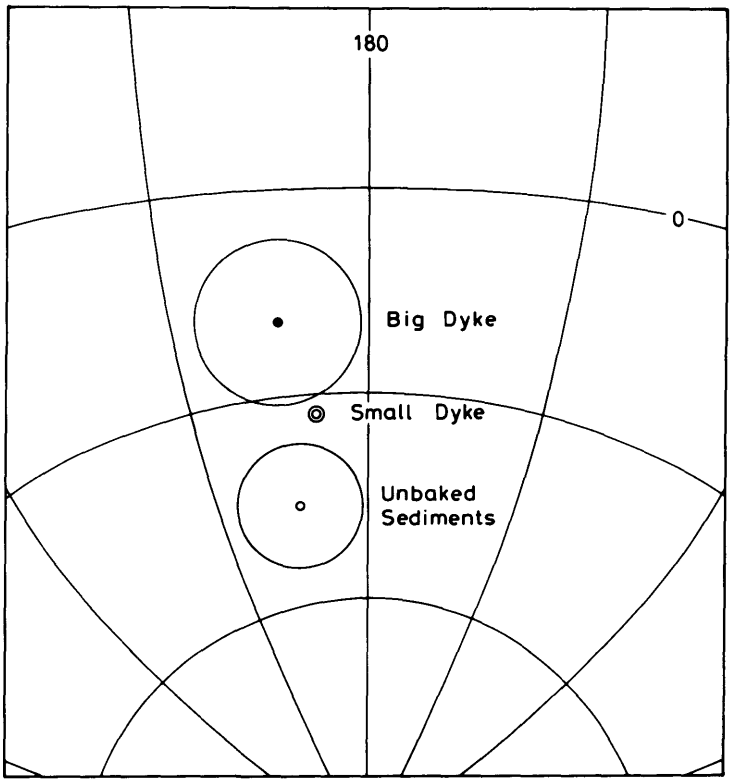


FIG. 3. Mean directions of the Big and Small Dykes, and of the unmetamorphosed Sediments.

The plot is on an equatorial stereographic projection, centred on a declination of 180°, with a 30° grid spacing. The Big Dyke has a positive inclination, marked by a solid dot, while the Small Dyke and sediments have negative inclinations, their mean directions being marked by a small circle. The circles around the mean direction correspond to α_{95}.

likely that the heating of the aureole was not simple thermal conduction, but involved heating by metasomatic fluids. Previous studies of Tertiary dykes in Northern England have suggested that the width of the thermal aureole often relates to the permeability of the host rocks (Thomas 1969). If such models are valid, then it seems likely that both dykes could have cooled rapidly and their magnetic remanences would then represent a spot reading in time of the geomagnetic field, i.e. the geomagnetic field direction over a period of some 20 years or so, although the Big Dyke is more likely to have retained an average direction acquired over 10^{2-4} years,and is thus more likely to have averaged out geomagnetic secular variations.

The clarity of the three different components within the metamorphic aureole

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of the Small Dyke appears to allow an evaluation of the sequence of remanence acquisition in this area. The presence of an "unmetamorphosed" component above 460°C, and the "Big Dyke" component at lower temperatures suggests that these sediments have been heated by the Big Dyke to about 450° prior to the intrusion of the Small Dyke, which only heated those within 1 m of it to 200°C. The discrete nature of the Small Dyke component within its aureole further indicates that this component is of thermal origin, since a chemical remanence, such as that associated with metasomatic fluids, should result in a range of new grain sizes and hence a range in blocking temperatures. It thus appears that the unmetamorphosed sediments carry a direction, of probable chemical origin, related to diagenesis in Lower Carboniferous times. The Big Dyke gave rise to a thermal remanence, probably during late stage diagenesis of the Lower Carboniferous sediments. The Small Dyke component was then superimposed on the two pre-existing components at some later time, probably in the Lower Permian.

Unfortunately, these acquisition ages are not consistent with their magnetic ages indicated by comparison with other Permo-Carboniferous rocks (Table 3), with the exception of the Big Dyke. The Big Dyke direction (194°, 19° uncorrected) is consistent with an upper Lower Carboniferous age, the corresponding pole being at 23°S, 341°E, and is little affected if corrected for the regional tilt (196°, 15° = 25°S, 340°E). The unmetamorphosed sediments and Small Dyke directions are similar to each other and suggest a much younger age, late Permian or even Triassic. The strong alteration of the Small Dyke suggests that its magnetization is more likely to be of chemical than thermal origin, despite the thermal discreteness of the components in its aureole. Both the Small Dyke and sediments could thus have been magnetized at a similar time as a result of

TABLE 3

Dunbar mean directions and pole positions.

	Dec	Inc	Pole position	
	Dunbar		S.	E.
L. Carb. Seds.	194	−46	60	332
Big Dyke	194	+19	23	341
Small Dyke	189	−33	51	344
North Western European Data (Tarling, 1983)				
Jurassic	182	−52	67	335
Triassic	191	−38	55	340
U. Permian	188	−21	44	346
M. Permian	191	−12	39	343
L. Permian	189	−14	41	346
U. Carboniferous	191	−12	39	343
M. Carboniferous	183	+10	29	354
L. Carboniferous	220	+26	13	318

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Permian weathering, although neither the Big Dyke nor its aureole show any signs of such later chemical remanence.

One of the standard geological tests for the age of magnetization, the contact test, thus appears to be fulfilled as the directions in the Small Dyke and its contact are identical, and differ from those of the country rocks, yet the pole positions and the degree of alteration of the Small Dyke suggest that this is invalid. The Big Dyke certainly appears to be of Lower Carboniferous rather than Permian age, confirming the evaluation by Francis (1975). However, this then requires an explanation of how the country rocks and Small Dyke were later altered, without affecting the Big Dyke remanence. Two possible explanations are that (a) the contact test is valid, although only positive at low temperatures, and the Big Dyke direction is simply a spot reading of the Permian geomagnetic field, i.e. displaced by some 40° from the average Permian direction by secular variations. The unmetamorphosed sediments would then have a Lower Carboniferous direction which is only coincidentally similar to the Permian geomagnetic field direction. Alternatively (b) the Permo–Triassic oxidation may have been inhibited in those rocks heated by the Big Dyke, which may have had their permeabilities reduced, and hence be less susceptibile to chemical alteration on both long and short time scales. However, similar arguments should hold for the Small Dyke. Thus neither explanation can be regarded as satisfactory.

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